

Money Movement fraud detection

What we built, what the data proves, and what still needs work

Status: A working dataset-driven research baseline is in the project. It is not a production fraud detector and does not yet validate every Money Movement typology.

1,323,234	1,719	2
uploaded transfers	fraud-labeled transfers	observed alert types

What the uploaded data contains

The supplied files are CSVs: transactions, alerts and accounts. The alerts call 936 labeled transfers cycle and 783 fan_in. The dataset does not include a cash-out channel, device/IP identifiers, currency, or the duration represented by one integer timestamp step. Its exact source is unconfirmed.

Evidence boundary: We can measure performance against the transaction fraud labels and the observed fan-in/cycle alerts. We cannot claim real-data detection quality for rapid cash-out, device sharing, or dormant mule activation.

Work completed across the four project layers

Layer	Delivered work
1. Attack intelligence	MM variant knobs and research notes; additive taxonomy and MM contract.
2. Synthetic world	Raw account-to-account transfer simulation, legitimate lookalikes, four attack injections, and validation report.
3. ML pipeline	CSV adapter, causal six-signal engineering, 2,048-D HDC, XGBoost, behaviour and anomaly scores, chronological evaluation.
4. Product workflow	Observed-edge graph, MM response path, dataset API, replay, Dataset Lab, generator, investigation view, and red-team panel.

The previous MM XGBoost comparison could fall back to hardcoded numbers, and its legacy transfer endpoint had a transfer-ID error. The new run uses a persisted XGBoost model and the endpoint now accepts its existing request shape.

Measured model performance

Held-out period: 193,281 transfers, including 266 labeled fraud cases. Training and threshold selection used earlier time periods.

Model	Precision	Recall	PR-AUC	ROC-AUC
HDC	3.12%	15.04%	1.53%	84.63%
XGBoost	3.52%	31.20%	2.33%	93.47%
Fusion	3.71%	21.43%	2.05%	90.79%

Honest reading: The fusion has slightly higher precision, but XGBoost alone has better recall and PR-AUC. At this fraud prevalence, the detector would still send many legitimate transfers for review. ROC-AUC does not capture that workload.

How the score is produced

Each transfer is mapped to six signals using only current and earlier events: fan-out, fan-in, rapid pass-through, amount layering, shared device when available, and account dormancy. The final research score combines HDC (50%), XGBoost (20%), behaviour rules (15%), and an Isolation Forest anomaly rank (15%). Thresholds were selected on validation data; the held-out period was used only for reporting.

Because device data and a currency threshold are absent, those parts of the six-signal representation are degraded or unavailable. The HDC result is therefore not a separate proof of all four named typologies.

Synthetic test and red team

Check	Observed result	What it means
Synthetic holdout	Fusion PR-AUC 63.78%; recall 65.67%	Pipeline recognizes some injected patterns in its own simulated world.
Red-team sweep	50 rings × 4 variants × 5 perturbation levels	A stress test of the same simulator and saved synthetic model.
Max perturbation	0% evasion for V1-V3; 4% for V4	Not reliable evidence of robustness; dense graph cycles can over-escalate.

Do not mix these numbers: The uploaded dataset's fraud rate is about 0.13%; the synthetic test has a far higher injected-fraud rate. Its metrics are not an estimate of real-world performance.

Verification and next steps

What was checked in code and in the running application

- Backend: 18 tests passed. The attack registry validated. Dataset probe and an uploaded zero-shot job completed; the zero-shot result made no accuracy claim.
- API: MM scan, generator, results, stream, graph, rings and audit routes returned successfully. An unknown graph account returned found: false. The legacy six-signal transfer endpoint returned successfully.
- Frontend: Production build passed. Browser checks showed real saved metrics in Dataset Lab, 200 scored transfer replays, an observed graph in Investigate, generated raw events and the red-team table.
- Regression: ATO and other category preset routes returned successfully. Exact before/after response bytes were not captured, so byte-identical behavior is not verified.

What is still incomplete

Area	Current limitation
Real-data coverage	No verified cash-out or dormant-mule labels; unknown timestamp-step duration, currency and data provenance.
Graph actions	Observed edges only, but risk is not yet amount/recency weighted. Dense cycles may trigger broad BLOCK_CHAIN decisions.
Analyst narrative	Evidence-only SAR draft template; no live LLM narrative integration.
Red-team loop	No adversarial-retraining before/after comparison or measured legitimate false-positive burden.
UI upload	API upload worked; in-browser file-picker automation did not complete, so that interaction remains unverified.

Recommended next work

1. Confirm data provenance/license, one timestamp step's duration, currency, and the precise meaning of each fraud label.
2. Obtain independently verified cash-out/withdrawal events, longer account history, allowed device/IP data, and case-level links between transfers and mule networks.
3. Fix graph escalation, then evaluate false positives and recall at a fixed analyst review capacity on a later time period and, ideally, an independent source. Calibrate scores before any real hold or freeze.

Project references: tanishq/MM_TRAINING_GUIDE.md has reproduction commands and formulas. tanishq/MM_DELIVERY_REPORT.md has detailed verification, deviations and the full file manifest. tanishq/results/mm_results.json has the measured test output.

This document describes the project team's prototype. It does not imply Mastercard approval or deployment.